The learning oracle

A self-contained statement of the framework, as agreed on 2026-08-22

2026-08-22

Contents

1 What this document is **1**

2 The problem the framework solves **2**

3 Notation **2**

4 Recipes and traces **3**

5 The learning oracle, passive form **3**

6 Views **4**

7 The learning oracle, interactive form **5**

8 Parties and the two kinds of game **6**

9 The adversary class **7**

10 Cost, and the ratio between parties **8**

11 The price of the search **9**

11.1 The price of aiming 9

11.2 The price of skipping is recipe-relative 10

11.3 An experiment: the measured exponent of the search 10

12 The corpus, mapped onto the framework **11**

13 The backdoor game, as far as it is fixed **13**

14 What is deferred, and what remains open about the oracle **14**

1 What this document is

This document states, from nothing, the formal framework that the survey's formalization section will rest on. It assumes the reader knows game-based cryptography and machine learning, and

assumes no prior exposure to the framework itself. Every object is defined before it is used. Every claim is labelled as a definition, a fact with a proof, a proposal, or an open question.

Two earlier artifacts exist. The extraction games in `games.tex` are one instance of what is defined here, and Section 12 says where they sit. The survey's frame of four coordinates, access point, criterion, budget and baseline, is recovered in Section 8, so nothing already written is lost, and one coordinate, the budget, is replaced by something more specific.

2 The problem the framework solves

A security definition for a learned model has to say what the adversary may do. For black-box access this is manageable: the adversary sends inputs and receives outputs, and one counts the queries. For white-box access it is not. The adversary holds the weights, and what it can learn from them depends on how training shaped them. A definition that covers that case seems to require a theory of the training map, the function from a corpus to a model. The map itself is public code that anyone can run. What no theory describes is what the map produces, and whether anything it produces can be obtained more cheaply than by running it.

Cryptography faced the same shape of problem with hash functions. No theory describes what SHA-256 does to its input, so the random oracle model replaces the hash function with an oracle that every party queries and no party can simulate, and proves theorems relative to that oracle. The generic group model does the same for group structure, and the Random Language Model of Brzuska and Couteau does it for language membership [9]. In each case the function is public, the hardness lies in reaching a *specified* output without paying the search or the walk, and the cost of an attack is counted in oracle queries.

The framework below makes that move for learning. The one object we cannot describe is the training map, so that is the one object the oracle idealizes. Inference is a computable function, analysis of weights is ordinary computation, and data is supplied by the game. The move has a precedent inside machine learning: the generic learning model of Villa et al. is this idealization used for one theorem, on the tension between a classifier's usefulness and membership privacy [46]. What is new here is using it as the ambient model every game lives in, and as the unit every party is priced in.

3 Notation

$\mathcal{X}, \mathcal{Y}$	input space and output space of a model
D	a corpus, a finite multiset of labelled examples from $\mathcal{X} \times \mathcal{Y}$
n	the security parameter, which scales the corpus, the model and the budgets
R	a recipe, Definition 1
w_t	the state of training after step t, a vector of weights
g_t	the gradient, or other per-step quantity, computed at step t
τ	a training trace, Definition 2
L	the learning oracle, Definitions 3 and 6
V	a view, Definition 4
$\mathcal{C}, \mathcal{A}$	the challenger and the adversary
steps	the cost unit, one step of the recipe

A *party* is any algorithm in a game. The challenger runs the game. The adversary tries to win it. Other named parties, an auditor or a second adversary, appear where a game needs them.

Everything is probabilistic polynomial time in n unless a game says otherwise.

4 Recipes and traces

The framework needs a name for a training procedure that is fixed and public but whose behaviour on a given corpus we do not claim to understand.

Definition 1 (Recipe)**.** A recipe R is a public description of a training procedure: an architecture, a loss, an optimizer, a batch rule, an initialization rule, a step count T, and any noise the procedure adds to its own computation. A recipe fixes a *step function* S that carries the current state w_{t-1} and a batch B_t to a per-step quantity g_t and a new state w_t.

A recipe is an object a paper would write in its experimental section. Stochastic gradient descent on a named architecture with a named learning rate is a recipe. Differentially private SGD, which clips and adds Gaussian noise to each gradient, is a different recipe, and the noise is inside it.

Definition 2 (Trace)**.** The trace of running recipe R on corpus D with coins r is the sequence

$$\tau = \big(w_0;\ (B_1, g_1, w_1);\ (B_2, g_2, w_2);\ \ldots;\ (B_T, g_T, w_T)\big),$$

where w_0 is the initial state drawn by the initialization rule, B_t the batch drawn by the batch rule, and $(g_t, w_t) = S(w_{t-1}, B_t)$. The final model is w_T. The *length* of the trace is T.

The trace is everything that happened during training, in order. A paper that releases a model releases w_T. A paper on gradient leakage studies some of the g_t. A proof-of-learning scheme asks a prover to exhibit a trace. These are all views of one object, which is the reason the trace and not the model is the oracle's output.

5 The learning oracle, passive form

Definition 3 (Passive learning oracle)**.** The passive learning oracle L_R for a public recipe R is an oracle that, on input a corpus D, draws fresh coins r, runs R on D with r, and returns the trace τ. It is the only source of traces in any game. Every party that holds a trace, or a model, or any part of a trace, obtained it through a call to L_R made by itself or by another party who then passed a view of it on. No party can compute a trace by any other means.

The last two sentences are the idealization, and they say something different from their random-oracle ancestor. A hash function is public, and so is the recipe. Nobody is prevented from running it. The oracle is a meter, not a keeper of a secret function: it runs the recipe so that the theory need not understand it, and it charges full price. What the idealization asserts is that the meter cannot be cheated, in two ways.

No skipping. The content of a T-step trace costs T steps. A hash chain of length T is computable by anyone, and by assumption not in fewer than T sequential calls. The oracle prices the walk to a converged model the same way. Sequential functions are the fine-grained home of this shape of assumption [28], with one boundary. They price depth, where the step count here is total work. Proof-of-learning, whose security rests on the unproved claim that forging a training record costs as much as training [32], is an instance of this postulate.

No aiming. A hash is inverted only by searching over inputs. Likewise there is no inverse map from desired model properties to weights. A party that wants a model with a specified behaviour,

one that agrees with a victim model, contains a backdoor, or forgets a record, chooses corpora and ports and pays more training, because learning is the only search over model space the literature contains.

Both postulates are one clause read from two sides. The honest party's work cannot be compressed, and the adversary's search has no primitive better than the oracle.

Two levels, kept apart. Inside the model the postulates are not assumptions. Cost is defined as metered steps and traces exist only through the oracle, so no in-model adversary can violate either postulate, and every theorem proved in the model is unconditional there. What the postulates name is the *transfer hypothesis*: a theorem proved in the model applies to a concrete system running a concrete recipe exactly when no procedure for that recipe class beats the meter. This is the random oracle methodology's own structure, and its known failure mode bounds what transfer may promise. Canetti, Goldreich and Halevi construct schemes secure in the Random Oracle Model with, in their words, "no implementation" that is secure [13]. Their counterexamples are contrived by design, and the analogous question here, whether a game secure in the learning-oracle model can fail for every concrete recipe, is open and is listed at the end of this document. Until it is answered, a theorem in this model is a theorem about generic adversaries, and its application to a deployed system is an argument, not a corollary.

Three consequences follow directly from the definition.

Fact 1 (Inference is free of the oracle)**.** *Given a model w_T, evaluating it on an input, computing its gradient with respect to an input, and any analysis of its weights are ordinary computations the party runs itself. None is an oracle call.*

This is what makes white-box access expressible. The adversary holding w_T may run singular value decomposition on a layer, probe features, or do anything else with the weights. The restriction is on how it *obtains* trained models, not on what it does with them.

Fact 2 (Fresh coins)**.** *Two calls to L_R on the same corpus return two independent traces. A party wanting several models trained on D makes several calls.*

Fact 3 (Cost)**.** *A call on a corpus D with recipe R costs T steps, where T is the recipe's step count, and submits $|D|$ examples. A party's oracle cost is the total number of steps it has caused to be run, and its total submitted corpus size, over all its calls.*

Steps rather than calls are the unit so that the passive oracle and the interactive oracle of Section 7 are charged alike. A shadow model trained from scratch costs T steps. Fine-tuning for T' steps from a held model is a recipe with step count T' whose initialization rule loads that model, and costs T'.

6 Views

A game rarely hands a whole trace to a party. It hands over what the scenario says the party sees, and that is a function of the trace.

Definition 4 (View)**.** A view V is a publicly known function of a trace, possibly randomized and possibly lossy. A game assigns each party a view. When a trace τ is produced, the party receives $\mathsf{V}(\tau)$ and nothing else of τ.

Views are game parameters, not oracle features. The oracle always produces the full trace. This is deliberate: if the oracle returned noisy gradients in one game and summary statistics in another it would be a different oracle per scenario, and there would be no single object to prove things about.

Example 1 (Views in use)**.**

- **Final model.** $\mathsf{V}(\tau) = w_T$. The released-weights setting.
- **Black-box access.** $\mathsf{V}(\tau)$ is a stateful wrapper that answers evaluation queries on w_T and counts them. The party never holds w_T. This is the API setting, and the count is the query budget.
- **Gradients without initialization.** $\mathsf{V}(\tau) = (g_1, \ldots, g_T)$. The party sees every gradient and not w_0.
- **Noisy channel.** $\mathsf{V}(\tau) = (g_t + \xi_t)_{t \le T}$ for independent noise ξ_t drawn by the view.
- **Sampled steps.** $\mathsf{V}(\tau) = (g_t)_{t \in I}$ for a random subset I of steps drawn by the view.
- **Summaries.** $\mathsf{V}(\tau)$ is the per-step mean and variance of the gradients.
- **Nothing.** $\mathsf{V}(\tau) = \bot$. The party learns that a trace was produced.

Fact 4 (Gradients plus initialization determine the model)**.** *For a deterministic step function, w_T is a function of w_0 and $(B_t, g_t)_{t \le T}$. Hence a party whose view contains the initial state and every per-step quantity holds the final model, and a scenario in which a party "sees the gradients but not the model" is only meaningful when its view withholds something.*

Proof. $w_t = S_2(w_{t-1}, B_t)$ for each t, where S_2 is the state coordinate of the step function, and the composition from w_0 is determined. □

Fact 4 is why the view is a named object. It forces a scenario to say what is withheld rather than to say informally that gradients were observed.

Recipe noise against view noise. Differentially private SGD adds noise inside the recipe. The learner trains on noisy gradients and the final model reflects it. A noisy channel adds noise inside a view. The learner trains on clean gradients and only the observer's copy is noisy. These are different objects. A theorem about one says nothing about the other, and a scenario must say which it means.

7 The learning oracle, interactive form

The passive oracle runs a whole training in one call with every part of the procedure honest. Federated learning, split learning and any setting where a party takes part in training while it runs are not of that shape. The party sends something into training at each round and receives something back, and it may deviate from what the recipe expects of it. The interactive oracle exposes the recipe's step as the unit of interaction.

Definition 5 (Ports)**.** A recipe may name *ports*. A port is a position in the step function where a quantity enters from outside or leaves to outside. Each port has an honest rule, the computation the recipe performs there when no external party is attached, and a *validity predicate*, the set of values the recipe accepts at that port.

Example 2 (Ports in two settings)**.** In federated averaging, each client is a port. Its outgoing quantity is the current global model, its incoming quantity is an update, its honest rule is local training on the client's data, and its validity predicate is whatever the server checks, for instance a norm bound. In split learning, the cut between the client half and the server half is a port in each direction. The client sends activations at the cut and receives gradients at the cut. The honest rule on each side is the forward or backward pass of that side.

Definition 6 (Interactive learning oracle)**.** The interactive learning oracle $\mathsf{L}^{\circ}_{\mathsf{R}}$ for a public recipe R with ports holds a training state. A game *attaches* parties to ports before training starts. The oracle then runs one step per call. In each step it sends to every attached party the outgoing quantities of that party's ports, waits for the incoming quantities, applies the validity predicates, substitutes the honest rule at any port that is unattached or whose incoming value is invalid, and advances the state. The trace records the honest and the received values at every port. As in Definition 3, the oracle is the only source of traces.

Attaching a party to a port gives it two things at once. It *reads* the port's outgoing traffic, which is a view of the trace restricted to that port. And it *writes* the port's incoming traffic, which is an intervention in the trace. A party that should only observe is attached to a port with an honest rule it cannot override, which is a read without a write, and is the same as giving it the corresponding view of a passive trace.

Fact 5 (The passive oracle is a special case)**.** *Running* $\mathsf{L}^{\circ}_{\mathsf{R}}$ *for* T *steps with no party attached produces the same distribution over traces as one call to* L_{R}*, and costs the same* T *steps.*

Definition 7 (Write budget)**.** A party attached to a port has a write budget: the number of steps at which it may send a value that differs from the honest rule, and a bound on how far the value may differ, stated in the port's own units. A write budget of zero is read-only attachment.

Write budgets are the second kind of budget in the framework. The first kind, from Fact 3, is what a party spends on the oracle, a read budget. The second is what a party is permitted to change, a write budget. A poisoning fraction is a write budget at the corpus. A malicious federated client's permitted deviation is a write budget at its port. The prior systematization of this area separates adversaries by whether they can read or also write [38]. Here that separation is two numbers.

8 Parties and the two kinds of game

With the oracle in place, a game is a protocol among parties that all share the same oracle. The shape that recurs across the corpus is the following.

Definition 8 (Game skeleton)**.** A game names the following.

1. A recipe R and a distribution over corpora, from which the challenger draws.
2. For each party, its attachment to ports and its write budget (Definitions 5 and 7), and its view of any trace the challenger produces (Definition 4).
3. For each party, a read budget in steps and in submitted examples (Fact 3).
4. An optional *secret*, a value drawn with min-entropy at least h, given to some parties and withheld from others.

5. A *criterion*, of one of the two kinds below.

6. A *baseline*, the value of the criterion achieved without access, or without the secret, or by the best strategy that ignores the trace entirely.

The survey's frame had four coordinates: access point, criterion, budget, and baseline. Items 1 and 2 are the access point, now as attachments and views. Item 5 is the criterion. Items 3 and 4 split the old budget into a read budget and a write budget, and add the secret, which the old frame lacked and which Section 13 needs. Item 6 is the baseline.

Definition 9 (Distinguish criterion)**.** The challenger prepares two *worlds*, two settings of the corpus or of the ports that differ in a stated way. Either a party submits the two and the challenger flips a coin b to pick one, or the challenger draws both. The challenger runs the oracle in world b, hands each party its view, and a designated party outputs a guess b'. Its advantage is $|\Pr[b' = b] - \frac{1}{2}|$.

Definition 10 (Produce criterion)**.** The challenger runs the oracle once, hands each party its view, and a designated party outputs a value. The party wins if the value satisfies a stated predicate within a stated tolerance. Its advantage is its winning probability minus the baseline.

The two criteria are the search and decision forms that cryptography has always had. Membership inference, differential privacy and undetectability are distinguish games. Reconstruction, extraction and forgery are produce games. A backdoor game is both at once, and Section 13 composes them.

Fact 6 (Worlds must differ)**.** *In a distinguish game, if the two worlds induce the same distribution on the designated party's view, its advantage is zero for every strategy, and indistinguishability holds for every bound on every party. A distinguish game is therefore only informative when the game states how the worlds differ and the statement is checkable.*

Proof. The party's output is a function of its view and its coins, and both have the same distribution in either world. □

Fact 6 is the analogue of requiring the two challenge messages in an indistinguishability game to be distinct. It matters for the backdoor game, and Section 13 returns to it.

9 The adversary class

Definition 11 (Learning-oracle algorithm)**.** A learning-oracle algorithm is an algorithm that obtains traces, models and training quantities only through the learning oracle, and otherwise computes freely. Its cost is its read budget, Fact 3, plus its own running time.

This is the adversary class the framework prices. The name in the rest of the document is *generic*, by analogy with generic group algorithms, which use the group only through its operations.

Definition 12 (Structural attack)**.** An attack is structural if it is not a learning-oracle algorithm, that is, if at some point it produces a model, a gradient or a trace without a corresponding oracle call.

The distinction is relative to what the oracle exposes, and two cautions fix its meaning.

First, an attack that uses only oracle calls but exploits a property of the models the recipe produces is generic. Carlini, Jagielski and Mironov extract the parameters of a ReLU network from real-valued outputs alone, by reading the network's piecewise linearity off second finite differences

of those outputs [16], and Canales-Martínez et al. make the extraction polynomial in depth [12]. Both use nothing but evaluations of the released model. Both are generic attacks on a structured model class, in the way a generic group algorithm may exploit that the group is cyclic. A generic-model lower bound for extracting ReLU networks must therefore sit at or below their query counts, and the model is calibrated by them rather than refuted.

Second, what the adversary may exploit is fixed by the game's *declaration*, a named object with three parts: the class the recipe's outputs lie in, the target family with the unit that prices it, and the noise the data carries. Noise is a member and not a detail, because noiseless problems admit algebraic escapes, parity by Gaussian elimination among them, and the surviving positive low-degree implications hold only for noisy problems [31]. What the adversary may exploit is fixed by the first part of the declaration. A game that declares the recipe outputs ReLU networks of depth h admits the attacks above. A game that declares nothing beyond "a function" admits less and proves more, for fewer real systems. The declaration is the one place where knowledge of learning enters the framework. The oracle does not remove that knowledge. It confines it to one line of the game.

Proposition 1 (A structural attack witnesses a transfer failure)**.** *Let a game be secure in the model against every generic adversary of budget B, and let a concrete implementation of the game, with recipe* R*, be broken by a real attack whose resources correspond to B under the game's cost accounting. Then no generic adversary of budget B simulates the attack, and the attack therefore contains a procedure that obtains a trace, a training quantity, or a model with the winning property for recipe class* R *at real cost below the metered price. Such a procedure is, by definition, a violation of the no-skipping or the no-aiming transfer hypothesis for* R*, that is, a way to train or to search model space faster than the recipe.*

Proof. If some generic adversary of budget B produced the attack's outputs from the attack's inputs, running it in the model would win the game within budget B, contradicting the in-model theorem. So none does. The attack nevertheless wins, so at some point it holds an object, a trace segment, a per-step quantity, or a winning model, that no sequence of oracle calls within its budget accounts for. The procedure producing that object obtains training output below the metered price, which is what the transfer hypotheses assert does not exist for R. □

This is the form in which the win-win theorem of Degwekar, Nakkiran and Vaikuntanathan enters [22]: either the construction is secure against generic adversaries, or a training shortcut exists, and a training shortcut is a learning-theoretic result of independent value. The proposition is close to a restatement of the definitions. Its use is to fix what a generic-model lower bound promises: not that no attack exists, but that any attack beating the bound trains, or searches model space, faster than the recipe, which the literature at present does not know how to do for the recipes this framework is intended for.

10 Cost, and the ratio between parties

Each party's cost has three parts: steps caused, examples submitted, and running time. A security statement bounds the adversary's advantage for every adversary whose cost lies under a stated threshold. The statements the survey cares about most set the threshold relative to an honest party's cost. Fine-grained cryptography writes that relation as an exponent [9, 35]: the adversary is permitted up to the honest cost raised to $1+\delta$, and the question is the largest δ at which security holds. In this framework the honest cost is a step count, which is dimensionless, so raising it to a power is well defined. The three cautions in `games.tex`, Section 6, on the three incompatible

conventions for δ and on the ceiling of Barak and Mahmoody being a query bound [6], apply unchanged.

One consequence is specific to the oracle. Every bound proved here is a bound on oracle steps, not on time. Barak and Mahmoody's adversary against Merkle puzzles makes few oracle queries and runs efficiently only in a relativized world where P = NP [6]. The same will hold of any adversary exhibited by a lower-bound argument in this model. A theorem here should say "steps" where it means steps.

11 The price of the search

The postulates of Section 5 say that the meter cannot be cheated. This section says what the meter reads, by grounding each postulate in the sharpest current results, and it reports one experiment. Everything here carries its epistemic status inline, because the ground shifts: two of the units this section could have been built on two years ago are now known to be algorithm-relative artifacts.

11.1 The price of aiming

The no-aiming postulate says the only search over model space is learning. Its concrete form is a lower bound on the oracle cost of any generic adversary that must produce a model with a target property, and the bound depends on the query type the recipe's steps realize.

For recipes whose steps are statistical queries, the floor is unconditional. A hidden k-sparse parity requires $d^{\Omega(k)}$ statistical queries at any useful tolerance, and this is the number Barak et al. describe SGD as approaching [5]. One hypothesis connects that floor to gradient recipes, and its status must be stated. A fresh-batch gradient step of batch size B evaluates a correlational statistical query to tolerance on the order of $B^{-1/2}$, up to concentration, so a T-step fresh-batch recipe reads as a correlational-query algorithm with T queries at that tolerance. The reading is standard and is how Barak et al. compare SGD to the SQ limit [5], and it is not a theorem. Dandi et al. state the inheritance for reused batches as an expectation explicitly [20]. Every use of a query floor for a gradient recipe in this framework routes through this simulation and carries its status. For Gaussian single-index models the current sharp single number is the generative exponent k^*: any statistical-query algorithm that outputs a direction with nontrivial overlap needs $n \gtrsim d^{k^*/2}$ samples, unconditionally within the class [19]. One caveat travels with that citation and it runs in the safe direction: the authors state that their theorem primarily measures the detection threshold, and the statistical-query estimation threshold is open, so the true price of recovery can only be higher than the floor quoted here. For multi-index models the same quantity governs subspace recovery, with an unconditional estimation upper bound at the matching rate and a low-degree lower bound [18]. Any reading of the low-degree half as polynomial-time hardness must carry the refutation of the quasi-polynomial low-degree conjecture and its surviving positive counterpart [10, 31].

Two units this section does not use, and why. The information exponent prices a single algorithm, online SGD from a random start [2], and the leap's lower bound holds for correlational statistical queries only, with its SGD side an open conjecture [1]. Both thresholds collapse under batch reuse [20], and the collapse is not an anomaly: reuse reaches the information-theoretic rate for polynomial links [36], a learning-rate schedule induces the same transition without carrying any sample across iterations, though each sample is queried twice within its own iteration [45], and full-batch descent separates from one-pass descent, with the separation proved for a truncated variant of the target and provably absent for the plain one [34]. What survives this wave is the

generative exponent, whose defining invariance is exactly closure under the transformations reuse exploits [19]. Proven, all of it, in the classes named.

The structural escape calibrates the postulate rather than refuting it. Noiseless parity is solvable by Gaussian elimination from $O(d)$ samples, an algorithm that manipulates samples algebraically and is no statistical-query algorithm. In the framework's terms that is a structural attack, and parity is the case where the win-win of Proposition 1 is not hypothetical: the generic floor is $d^{\Omega(k)}$, the structural attack is linear algebra, and the gap between them is a named theorem about what learning-shaped computation cannot do.

11.2 The price of skipping is recipe-relative

The no-skipping postulate as stated in Section 5 prices the walk of one recipe. The literature now forces exactly that scoping, and the force is worth exhibiting, because a stronger reading of the postulate is false.

The walk's length is a property of the pair recipe and query type, not of the task. The batch-reuse and learning-rate results above shorten the same walk to the same target by changing the recipe slightly [20, 36, 45], and empirically the compute needed to reach a fixed language-model loss has halved roughly every eight months across a decade of recipe drift [29]. So a task-level no-skipping claim is refuted, and a recipe-level one is the strongest defensible form. Within a fixed recipe, the compute-to-loss frontier is a theorem in solvable models: for power-law random features under one-pass stochastic gradient descent the loss curve is exact and the compute-optimal frontier is a derived object [39], which is the theorem-shaped form of the empirical scaling laws [33, 30], whose fitted constants are themselves contested [7]. The mechanism literature makes the exponent explicitly algorithm-dependent: allowing feature learning nearly doubles the compute exponent on hard tasks [8], and a good data-pruning rule changes the law's form outright [41]. None of these results claims a lower bound over all algorithms, and no such bound exists. Open, and it is the open problem under every compute-pricing claim in this survey.

Two consequences are definitional in the oracle model and consistent with the empirical record. First, amortization is not skipping. Distillation reaches a target cheaper only when a stronger teacher already exists, and the teacher's training was an oracle call somebody paid for [11]. In the model, a party that builds a derivative model builds it from views of prior traces, so the cost sits upstream in the ledger. There is no free trace, only amortized traces. Second, the price of a view is not the price of the trace. Proof-of-learning verification accepts a lossy view of the trace, and the published forgeries attack that view, succeeding without paying the walk [24], while exact reproduction of a trace remains empirically out of reach [43]. A sound proof-of-learning is precisely a view whose price equals the trace's, and whether one exists is the open problem that literature keeps encountering from inside.

11.3 An experiment: the measured exponent of the search

The postulates predict that a generic party obtaining a model it cannot aim at pays a cost that grows as a power of the problem size, with the exponent set by the target class. Sparse parity is the cleanest instance, because the statistical-query floor is unconditional and the statistical baseline of $\Theta(k \log d)$ samples makes the gap between information and computation visible. Barak et al. report the phase transition and its scaling [5]. The run below reproduces the law in their regime with this framework's accounting, so that the survey's numbers are its own.

A two-layer ReLU network of width 100 is trained by minibatch SGD, batch 128, learning rate 0.01, on fresh samples per step, to learn a hidden 3-sparse parity of d bits, for d from 20 to 60, 8

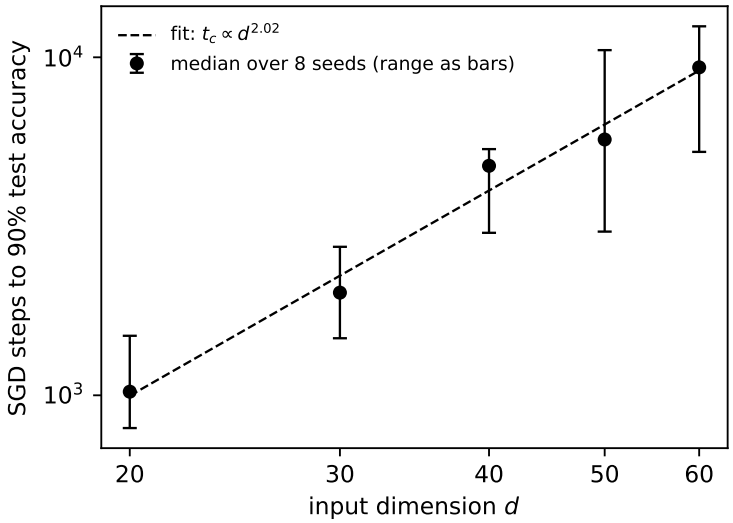


Figure 1: Median SGD steps to 90% test accuracy against input dimension for a hidden 3-sparse parity, log-log, 8 seeds per point, bars spanning the seed range. The fitted exponent 2.02 is the measured price of the search in the recipe stated in the text.

seeds per dimension. The cost recorded is oracle steps to reach 90% test accuracy. All 40 of 40 runs reached it. The median cost rises from 1025 steps at $d = 20$ to 9337 at $d = 60$, and the log-log fit across dimensions gives

$$t_c \propto d^{2.02},$$

Figure 1. Three sentences fix what the fitted exponent is and is not. It sits in the power-law family $t_c \propto d^{\alpha k}$ that Barak et al. document, whose exponent depends on the architecture, the batch and the regime, and whose own scaling plots show it worsening at larger d [5]. It is consistent with the query floor through the simulation above, since the floor constrains the product of queries and squared tolerance and fixes no step exponent at a given batch. It does not measure the floor, and no experiment can, since a floor quantifies over algorithms. The convergence times are heavy-tailed across seeds at fixed dimension, which matches the initialization sensitivity Barak et al. document. In a second regime, batch 8 with a larger learning rate, the wall is sharper than any fit: the median at $d = 30$ is 6940 steps and none of nine runs at $d \in \{60, 90, 120\}$ reached threshold within 300000 steps, a greater than forty-fold increase against a doubling of dimension. Empirical, all of it, in the configurations stated.

What the experiment does and does not show. It shows one recipe's cost growing as the power law the class floor predicts, in the regime where the recipe's steps are noisy correlational queries. It does not measure the floor itself, since no experiment can quantify over algorithms, and it says nothing about structural adversaries, for which noiseless parity is easy. It is the empirical face of the no-aiming postulate for the class that trains, and nothing more.

12 The corpus, mapped onto the framework

This section places the families the survey covers into Definition 8, to show that nothing in the framework is particular to one of them.

Membership inference. Distinguish. The worlds differ by one record in the corpus. The adversary's view is the final model, or black-box access. The baseline is the blind attack, which Das et al. show beats published attacks on eight benchmarks [21]. A shadow-model attack [40, 15] is generic and costs $N \cdot T$ steps for N shadow models. The verifier of Zhang et al., who cannot sample the null hypothesis because doing so "would require training new large models" [47], is an auditor with a read budget of zero steps.

Membership inference from gradients. Distinguish, with the adversary's view set to gradients, Example 1. Fact 4 says which lossy view the scenario must name.

Gradient inversion. Produce. The view is gradients or activations at a port, and the criterion is reconstruction within a tolerance. The reconstruction-robustness definition of Balle, Cherubin and Hayes already supplies the criterion and the oblivious baseline [3], and the gradient-inversion literature, for instance [4], supplies attacks with no stated criterion. Passive observation of a federated round is a read-only attachment. A malicious server is a write.

Computational differential privacy. Distinguish, with the worlds drawn by the challenger as neighbouring corpora and the distinguisher bounded [37]. Recipe noise, Section 6, is where the mechanism lives.

Evasion. Produce. The criterion is a label change inside a ball around a given input. The baseline is random noise in the same ball, which the concentration results show already succeeds on some tasks [26]. Access is a view: black-box is the metered wrapper, white-box is the final model, and transfer is a shadow model plus the adversary's own gradient computation. The number of gradient steps an attack is allowed, which the evaluation guidance of Carlini et al. declines to bound [14] and which Suya et al. use to rank attacks [44], is the adversary's running time in Definition 11.

Extraction. Produce. The games of `games.tex` are this case with the view set to black-box access, and their two-party pricing is Section 10.

Poisoning. A write budget at the corpus. Availability poisoning is produce, with the criterion an accuracy drop. Certified defences bound the drop for every write within the budget [42]. Robust estimation in high dimensions is the case where the defender is polynomial time and the writer is unbounded except in the fraction it may change [23].

Proof-of-learning. Produce. The criterion is a trace that a verifier accepts. The honest prover's trace comes from T oracle steps, and forging is producing an accepted trace in fewer. Jia et al. require that forging cost at least as much as honest training and conjecture rather than prove it [32]. In this framework the requirement is a bound on the forger's steps relative to T, and the attacks that break their scheme [24] are generic or structural according to whether they call the oracle.

Watermarking. Distinguish, with a secret as the hinge: the detector holds the key and the distinguisher does not, as in Christ, Gunn and Zamir [17].

Filtering and auditing. Distinguish, where the designated party is an auditor whose read budget is stated relative to the model's own cost.

13 The backdoor game, as far as it is fixed

This is the one game that composes both criteria. It is stated here to the extent it has been agreed, with the open decisions marked.

Fixed. Three roles. A *planter* intervenes during training through a write budget at a port or at the corpus. An *auditor* observes through a view and a read budget and outputs a world guess, the distinguish criterion. An *activator* holds a secret drawn with min-entropy h, chosen by the planter, and tries to elicit a stated behaviour from the final model, the produce criterion. The activator's baseline is the success of an activator without the secret, which is the non-replicability property of Goldwasser, Kim, Vaikuntanathan and Zamir [27]. The auditor never holds the secret, because an auditor given the secret detects by running the activator, so undetectability is meaningful only against auditors without it.

Fixed. The planter is generic, Definition 11. The constructions in the corpus that achieve undetectability all give the planter control of training itself and all use cryptographic structure inside the model, digital signatures, continuous LWE, sparse PCA, obfuscation, or a hash circuit compiled into a transformer [27]. A planter that only writes to the corpus and faces an honest recipe is the case those constructions do not cover.

Fixed. The auditor performs detection, not reconstruction. The question is whether some intervention occurred, and the auditor is not asked to say which.

Open 1 (Non-triviality). By Fact 6, a planter whose writes leave the distribution of the auditor's view unchanged is undetectable and has done nothing. The game needs a condition on world 1 that rules this out. The activator's success is one such condition. If the game is to be stated without requiring the backdoor to work at inference, another condition must replace it, for instance a minimum write budget together with a declared difference in the label distribution of the written examples. Which condition to use is not decided.

Open 2 (Where the planter writes). The planter may write at the corpus, at a training port, or by replacing the data distribution itself. These give different games, and the last puts no fraction in the game at all. Which of them the survey's backdoor section treats first is not decided.

The corpus-level half of the game's first theorem is a specialization of existing machinery, and it is worth stating because it prices a party this literature leaves unpriced.

Proposition 2 (Undetectability against a query-bounded corpus auditor)**.** *Fix the backdoor game in which world 0 draws the corpus from a distribution D_0 and world 1 draws it from a distribution the planter selects out of a family $\mathcal{S}$, and fix the auditor's view to be q queries to a* $\mathrm{VSTAT}(m)$ *oracle on the corpus distribution. If the statistical dimension* $\mathrm{SDA}(D_0, \mathcal{S}, \bar{m})$ *of Feldman et al. is at least q' at the discrimination parameter matching m, then no auditor with $q < q'$ queries decides the world with the success their framework fixes, with parameters inherited from their lower bound [25]. In particular, for planted bipartite k-clique with $k = o(\sqrt{n})$, an auditor restricted to* $\mathrm{VSTAT}(o(n^2/k^2))$ *needs $n^{\Omega(\log n)}$ queries.*

Proof. The auditor's task, as fixed, is the decision problem between D_0 and $\mathcal{S}$ in the VSTAT model, which is the problem the statistical-dimension lower bound of Feldman et al. prices, and the planted bipartite clique instantiation is their main theorem [25]. Nothing beyond the specialization is claimed. □

Three boundaries keep the proposition honest. It prices an auditor whose view of the corpus is statistical averages, and says nothing about an auditor who reads individual records. It does not touch the auditor who views the trained model rather than the corpus, and that transfer is open. And it does not assert that a generic planter can simultaneously meet a non-triviality condition with a clique-shaped poison, which is Open 1.

Open 3 (The model-level auditor). Whether Proposition 2 transfers to an auditor viewing the final model, rather than the corpus, is not known, and it is the half of the first theorem that would price backdoor detection as deployed defences practice it.

14 What is deferred, and what remains open about the oracle

Deferred 1 (The criterion for generation). The produce criterion assumes a predicate the challenger can evaluate. For a language model asked to be harmless, the predicate is not something we can write down. Villa et al. idealize that predicate as a ground truth and bound the gap to a computable judge [46]. This framework does not yet include that second idealization, and a game about generation needs it.

Deferred 2 (Classification). The tolerance in a produce criterion assumes a metric on outputs. For classifiers under the zero-one metric, the propositions in `games.tex` that rest on a Lipschitz constant do not apply, and the produce games need a separate treatment.

Open 4 (Declaration). Section 9 fixes that what the adversary may exploit is what the game declares about the recipe's output. The framework does not yet say what a declaration may contain. Too little, and every theorem is about random functions. Too much, and the declaration is the theory of learning the oracle was introduced to avoid. The declaration must also assert that the recipe has no known shortcut, because the postulates are false where one exists: a convex recipe such as linear regression has a closed-form endpoint, so the content of its trace is obtainable with none of its steps. The idealization is only credible for recipe classes in the no-known-shortcut regime, as the random oracle model is only credible for hash functions with no known structure. Where that line sits is the main open question about the oracle itself.

Open 5 (Uninstantiability). Random-oracle proofs are known to have instantiations for which the proved theorem fails, by the construction of Canetti, Goldreich and Halevi [13]. Whether the analogous phenomenon arises when the learning oracle is replaced by a concrete recipe is not known. A learning analogue of their diagonalization, a game secure in this model that every concrete recipe implements insecurely, would put the same known bound on this methodology that theirs put on the random oracle.

References

[1] Emmanuel Abbe, Enric Boix-Adserà, and Theodor Misiakiewicz. Sgd learning on neural networks - leap complexity and saddle-to-saddle dynamics. arXiv:2302.11055, 2023.

[2] Gerard Ben Arous, Reza Gheissari, and Aukosh Jagannath. Online stochastic gradient descent on non-convex losses from high-dimensional inference. arXiv:2003.10409, 2021.

[3] Borja Balle, Giovanni Cherubin, and Jamie Hayes. Reconstructing training data with informed adversaries. In *IEEE S&P*, 2022.

[4] Mislav Balunović, Dimitar I. Dimitrov, Nikola Jovanović, and Martin Vechev. Lamp: Extracting text from gradients with language model priors. In *NeurIPS*, 2022.

[5] Boaz Barak, Benjamin L. Edelman, Surbhi Goel, Sham Kakade, Eran Malach, and Cyril Zhang. Hidden progress in deep learning - sgd learns parities near the computational limit. arXiv:2207.08799, 2022.

[6] Boaz Barak and Mohammad Mahmoody-Ghidary. Merkle puzzles are optimal — an $o(n^2)$-query attack on any key exchange from a random oracle. In *CRYPTO*, 2009.

[7] Tamay Besiroglu, Ege Erdil, Matthew Barnett, and Josh You. Chinchilla scaling: A replication attempt. arXiv:2404.10102, 2024.

[8] Blake Bordelon, Alexander Atanasov, and Cengiz Pehlevan. How feature learning can improve neural scaling laws. In *ICLR*, 2025.

[9] Chris Brzuska and Geoffroy Couteau. On building fine-grained one-way functions from strong average-case hardness. In *EUROCRYPT*, 2022.

[10] Rares-Darius Buhai, Jun-Ting Hsieh, Aayush Jain, and Pravesh K. Kothari. The quasi-polynomial low-degree conjecture is false. In *FOCS*, 2025.

[11] Dan Busbridge, Amitis Shidani, Floris Weers, Jason Ramapuram, Etai Littwin, and Russ Webb. Distillation scaling laws. In *ICML*, 2025.

[12] Isaac A. Canales-Martínez, Jorge Chávez-Saab, Anna Hambitzer, Francisco Rodríguez-Henríquez, Nitin Satpute, and Adi Shamir. Polynomial time cryptanalytic extraction of neural network models. In *EUROCRYPT*, 2024.

[13] Ran Canetti, Oded Goldreich, and Shai Halevi. The random oracle methodology, revisited. In *STOC*, 2004.

[14] Nicholas Carlini, Anish Athalye, Nicolas Papernot, Wieland Brendel, Jonas Rauber, Dimitris Tsipras, Ian Goodfellow, Aleksander Mądry, and Alexey Kurakin. On evaluating adversarial robustness. arXiv:1902.06705, 2019.

[15] Nicholas Carlini, Steve Chien, Milad Nasr, Shuang Song, Andreas Terzis, and Florian Tramèr. Membership inference attacks from first principles. In *IEEE S&P*, 2022.

[16] Nicholas Carlini, Matthew Jagielski, and Ilya Mironov. Cryptanalytic extraction of neural network models. In *CRYPTO*, 2020.

[17] Miranda Christ, Sam Gunn, and Or Zamir. Undetectable watermarks for language models. arXiv:2306.09194, 2023.

[18] Alex Damian, Jason D. Lee, and Joan Bruna. The generative leap - sharp sample complexity for efficiently learning gaussian multi-index models. arXiv:2506.05500, 2025.

[19] Alex Damian, Loucas Pillaud-Vivien, Jason D. Lee, and Joan Bruna. Computational-statistical gaps in gaussian single-index models. arXiv:2403.05529, 2024.

[20] Yatin Dandi, Emanuele Troiani, Luca Arnaboldi, Luca Pesce, Lenka Zdeborová, and Florent Krzakala. The benefits of reusing batches for gradient descent in two-layer networks - breaking the curse of information and leap exponents. arXiv:2402.03220, 2024.

[21] Debeshee Das, Jie Zhang, and Florian Tramèr. Blind baselines beat membership inference attacks for foundation models. In *DATA-FM workshop @ ICLR*, 2025.

[22] Akshay Degwekar, Preetum Nakkiran, and Vinod Vaikuntanathan. Computational limitations in robust classification and win-win results. In *COLT*, 2019.

[23] Ilias Diakonikolas, Gautam Kamath, Daniel M. Kane, Jerry Li, Ankur Moitra, and Alistair Stewart. Robust estimators in high dimensions without the computational intractability. arXiv:1604.06443, 2016.

[24] Congyu Fang, Hengrui Jia, Anvith Thudi, Mohammad Yaghini, Christopher A. Choquette-Choo, Natalie Dullerud, Varun Chandrasekaran, and Nicolas Papernot. Proof-of-learning is currently more broken than you think. arXiv:2208.03567, 2023.

[25] Vitaly Feldman, Elena Grigorescu, Lev Reyzin, Santosh S. Vempala, and Ying Xiao. Statistical algorithms and a lower bound for detecting planted cliques. In *JACM*, 2017.

[26] Justin Gilmer, Luke Metz, Fartash Faghri, Samuel S. Schoenholz, Maithra Raghu, Martin Wattenberg, and Ian Goodfellow. The relationship between high-dimensional geometry and adversarial examples (adversarial spheres). arXiv:1801.02774, 2018.

[27] Shafi Goldwasser, Michael P. Kim, Vinod Vaikuntanathan, and Or Zamir. Planting undetectable backdoors in machine learning models. arXiv:2204.06974, 2022.

[28] Jiaxin Guan and Hart Montgomery. On sequential functions and fine-grained cryptography. In *CRYPTO*, 2024.

[29] Anson Ho, Tamay Besiroglu, Ege Erdil, David Owen, Robi Rahman, Zifan Carl Guo, David Atkinson, Neil Thompson, and Jaime Sevilla. Algorithmic progress in language models. arXiv:2403.05812, 2024.

[30] Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, Tom Hennigan, Eric Noland, Katie Millican, George van den Driessche, Bogdan Damoc, Aurelia Guy, Simon Osindero, Karen Simonyan, Erich Elsen, Jack W. Rae, Oriol Vinyals, and Laurent Sifre. Training compute-optimal large language models (chinchilla). arXiv:2203.15556, 2022.

[31] Jun-Ting Hsieh, Daniel M. Kane, Pravesh K. Kothari, Jerry Li, Sidhanth Mohanty, and Stefan Tiegel. Rigorous implications of the low-degree heuristic. In *STOC*, 2026.

[32] Hengrui Jia, Mohammad Yaghini, Christopher A. Choquette-Choo, Natalie Dullerud, Anvith Thudi, Varun Chandrasekaran, and Nicolas Papernot. Proof-of-learning: Definitions and practice. In *IEEE S&P*, 2021.

[33] Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models. arXiv:2001.08361, 2020.

[34] Filip Kovačević, Hong Chang Ji, Denny Wu, Mahdi Soltanolkotabi, and Marco Mondelli. Full-batch gradient descent outperforms one-pass sgd - sample complexity separation in single-index learning. arXiv:2602.02431, 2026.

[35] Rio LaVigne, Andrea Lincoln, and Virginia Vassilevska Williams. Public-key cryptography in the fine-grained setting. In *CRYPTO*, 2019.

[36] Jason D. Lee, Kazusato Oko, Taiji Suzuki, and Denny Wu. Neural network learns low-dimensional polynomials with sgd near the information-theoretic limit. arXiv:2406.01581, 2024.

[37] Ilya Mironov, Omkant Pandey, Omer Reingold, and Salil Vadhan. Computational differential privacy. In *CRYPTO*, 2009.

[38] Nicolas Papernot, Patrick McDaniel, Arunesh Sinha, and Michael P. Wellman. Sok: Security and privacy in machine learning. In *IEEE EuroS&P*, 2018.

[39] Elliot Paquette, Courtney Paquette, Lechao Xiao, and Jeffrey Pennington. 4+3 phases of compute-optimal neural scaling laws. arXiv:2405.15074, 2024.

[40] Reza Shokri, Marco Stronati, Congzheng Song, and Vitaly Shmatikov. Membership inference attacks against machine learning models. arXiv:1610.05820, 2017.

[41] Ben Sorscher, Robert Geirhos, Shashank Shekhar, Surya Ganguli, and Ari S. Morcos. Beyond neural scaling laws, beating power law scaling via data pruning. In *NeurIPS*, 2022.

[42] Jacob Steinhardt, Pang Wei Koh, and Percy Liang. Certified defenses for data poisoning attacks. arXiv:1706.03691, 2017.

[43] Mohamed Suliman, Anisa Halimi, Swanand Ravindra Kadhe, Nathalie Baracaldo, and Douglas Leith. Towards a re-evaluation of data forging attacks in practice. arXiv:2411.05658, 2025.

[44] Fnu Suya, Anshuman Suri, Tingwei Zhang, Jingtao Hong, Yuan Tian, and David Evans. Sok: Pitfalls in evaluating black-box attacks. arXiv:2310.17534, 2024.

[45] Konstantinos Christopher Tsiolis, Alireza Mousavi-Hosseini, and Murat A. Erdogdu. From information to generative exponent, learning rate induces phase transitions in sgd. arXiv:2510.21020, 2025.

[46] Federico Villa, F. Betül Durak, Tadayoshi Kohno, Tapdig Maharramli, and Franziska Roesner. Extending the formalism and theoretical foundations of cryptography to ai. arXiv:2603.02590, 2026.

[47] Jie Zhang, Debeshee Das, Gautam Kamath, and Florian Tramèr. Position: Membership inference attacks cannot prove that a model was trained on your data. arXiv:2409.19798, 2025.